Couchbase CAPELLA

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The Operational Data Platform for AI

Couchbase Capella

March 2026

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Agenda

Why Capella?

Key Concepts & Architecture

Mobile App Services

AI Services

Security & Compliance

Pricing Model

Performance Benchmarks

Why Capella

>

Capella: Couchbase as a Service

Reducing your operational efforts to focus on your business

Database-as-a-Service

Fully managed database

- Maximize convenience
- Easy to start, manage, and scale
- Industry leading price-performance
- Highly available
- Robust secure

App Services

Sync, store, query, search, and analyze at the edge

- Offline first embeddable database
- Sophisticated, automated syncing
- Design for max speed and uptime
- Peer-to-peer sync
- Strong security, SQL and device support

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Couchbase Capella

Key capabilities

Fully-Managed Database

Enterprise-Grade Performance

Development Flexibility

High Availability

Automated Scaling In/Out & Up/Down

Single Pane for Multi-cluster, Multi-region

Mobile and IoT app sync & embeddable database

Security Peace of Mind

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Simply Your Operations and Costs

More value: Shift resources to more critical priorities

Couchbase CAPELLA

Couchbase Subscription

+

Customer: IT Resource Management Costs & Overhead

+

Cloud Provider: Infrastructure Costs

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Free up Your Team's Time for More Critical Tasks

Category	Tasks	Customer	Capella
Design work	Database design and sizing	Responsible	Prof. Serv.
Cloud and database Infrastructure	Setting up cloud infrastructure		Automated
	Configuring host OS and ongoing management		Automated
	Setting up networking		Automated
	Infrastructure and database monitoring		Automated
Database Management	Couchbase server settings	Minimal	Automated
	Security	Shared	Automated
	Replication across clusters	Customer configures	Automated
	Configure cloud security for database cluster		Automated
	Configure data encryption at rest and inflight		Automated
	DR: Failover and HA management		Automated
	Configure Couchbase services		Automated
	User management	Responsible	Automated
	Database backups	Sets schedule	Automated
	Database scaling and upgrades	Customer decision	Automated
	Rebalance nodes and cluster after changes		Automated
	Track node failovers		Automated
	Implement Couchbase best practices		Automated
	Log collection and upload to support		Automated
	Monitoring and recovering instances		Automated
	Alerts	Review & Action	Automated
AI Services	Model deployment and hosting	Choose	Automated
	LLM caching and guardrails	Configure	Automated
	Data ingestion, vectorization and indexing	Configure	Automated
	Agent tracking	Configure	Automated
	Process data with SQL++ and LLMs	Choose	Automated

Couchbase Drives Down Costs

53% ...of customers had >50% TCO reduction*

Performance at Scale

- Memory first architecture
- Optimize services to needs
- Dollar for dollar winner

Single Platform

- Reduce license & hardware
- Less to learn, code, integrate
- Leverage SQL skill sets

Cloud Advantages

- OPEX vs CAPEX
- Add servers as needed
- Centralized management

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*TechValidate: More stories here

Couchbase Capella iQ

Lowering the learning curve, accelerating development

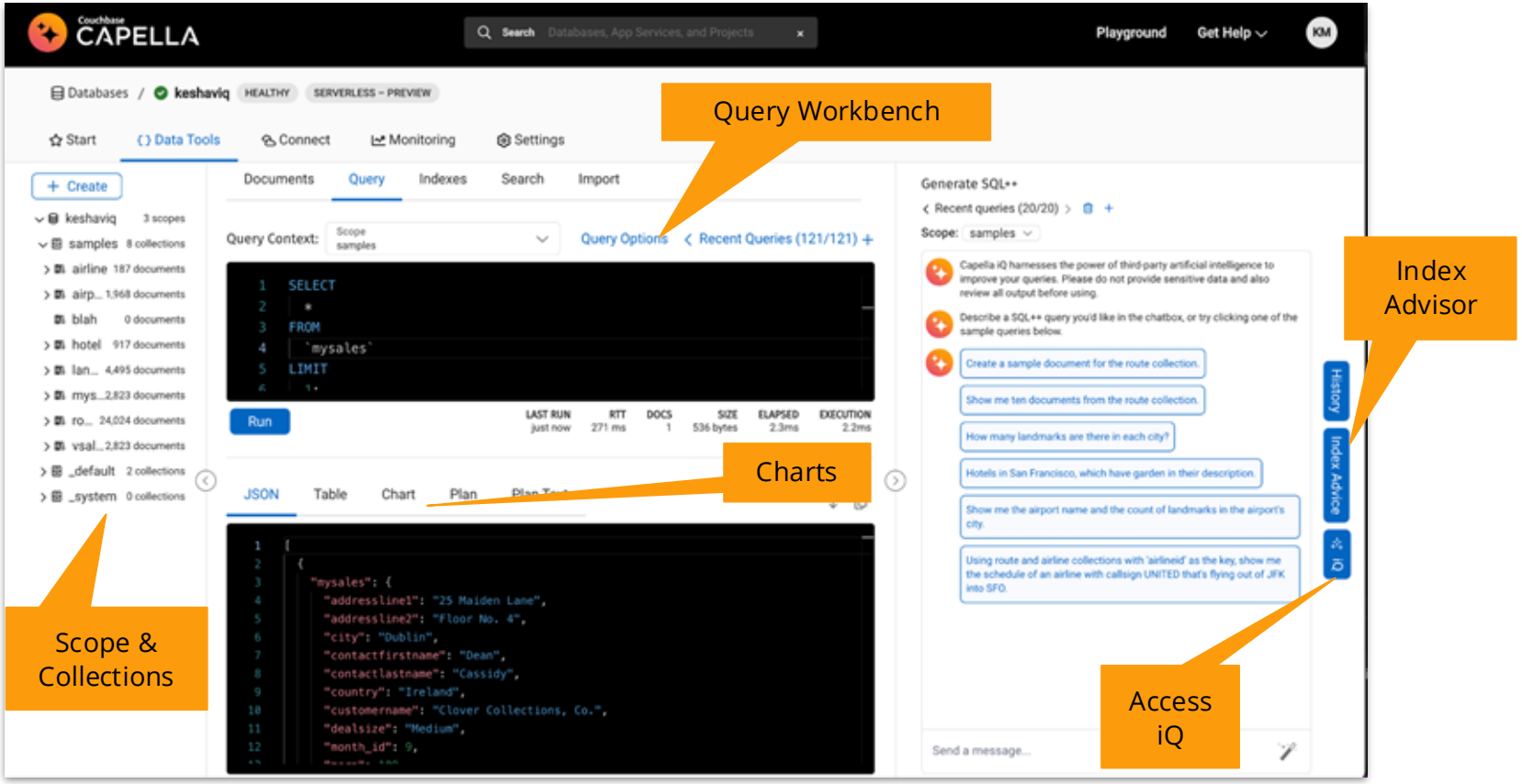
Interact via natural language

- Make automated code recommendations
- SQL queries
- Indexes
- Sample documents
- Application code

Add context to prompts

Users will pick their LLM

Leverage AI for building & testing



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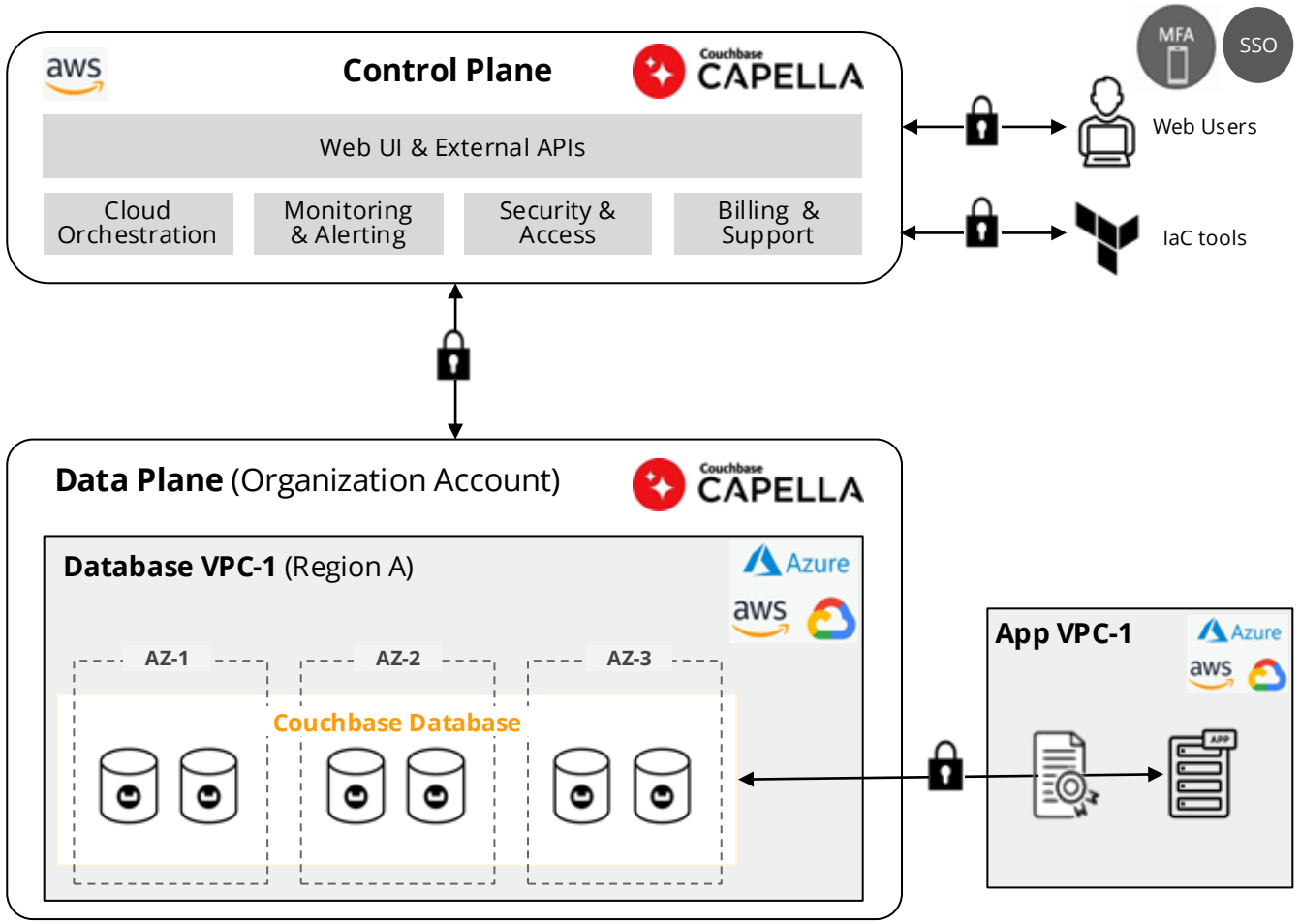
Key Concepts & Architecture

Architecture



- Control Plane
  - Manage all databases, a single pane
  - Web UI users access point
- Data Plane
  - On all 3 major clouds
  - Database = Cluster
- Applications
  - Connect directly to databases
    - Over public connection
    - VPC peering
    - PrivateLink
  - All communications are encrypted

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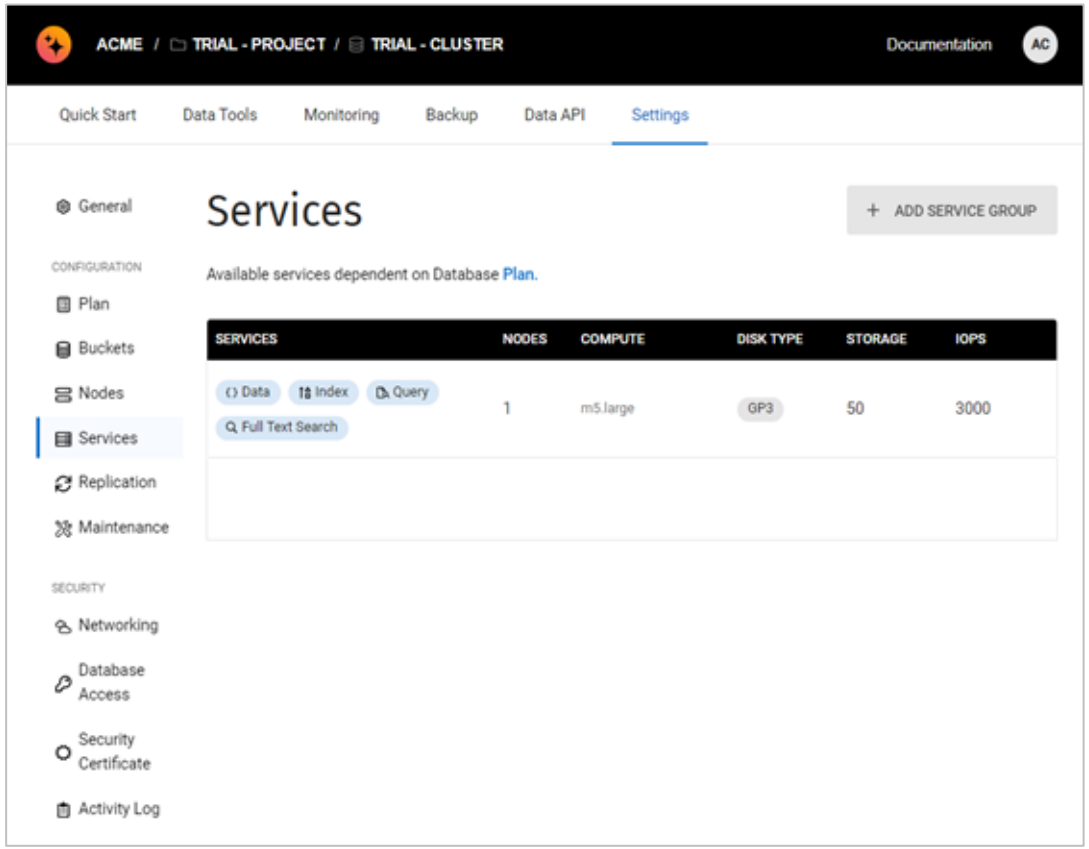
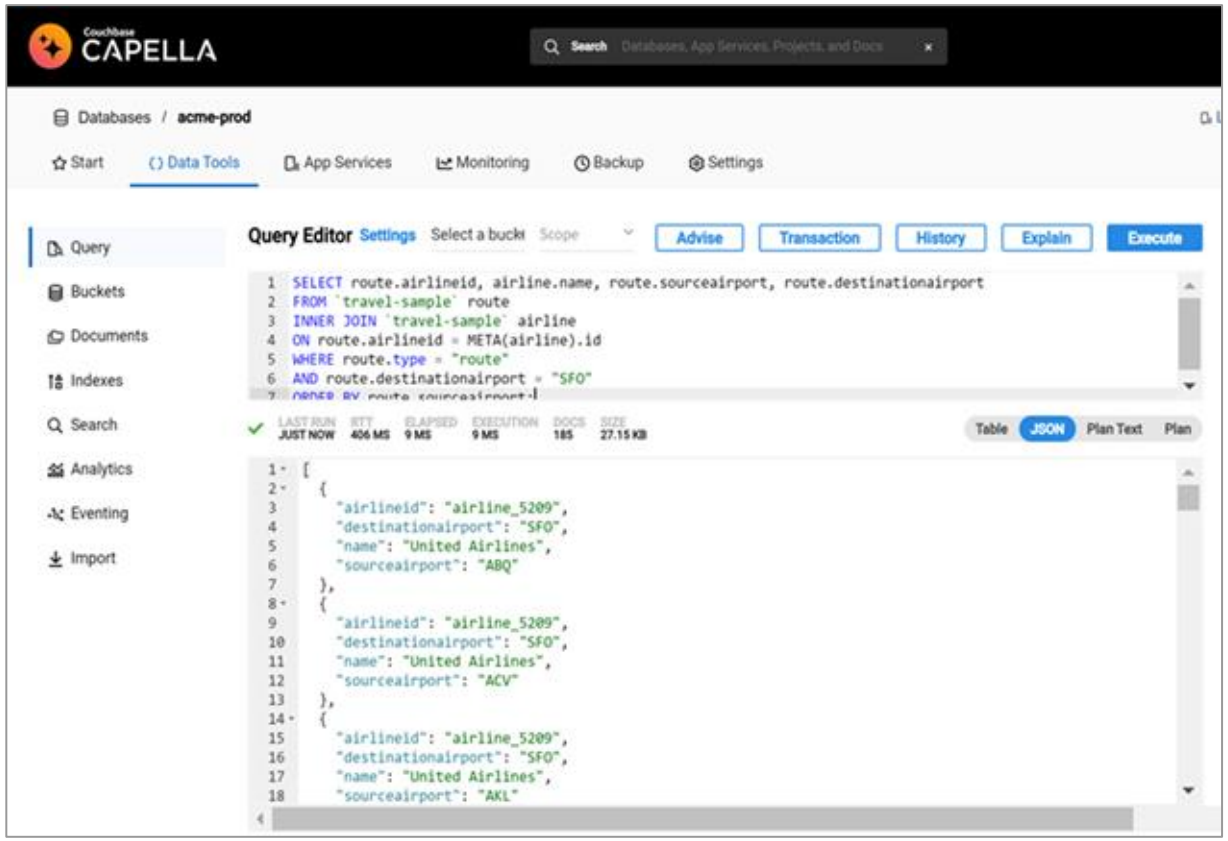
Control Plane

User Interface

Administration



Data Tools



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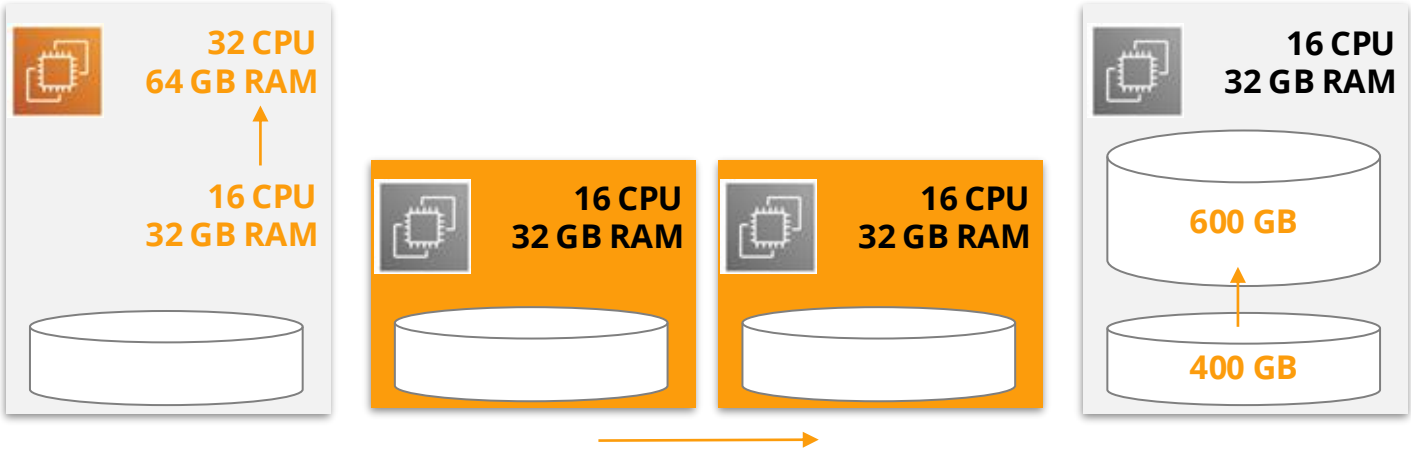
Automated Scaling

Align CPU, RAM and Storage to your Workload

Scale Vertically
CPU/RAM

Scale Horizontally
Number of nodes

Scale Storage
Disk size



- Compute
  - 20 EC2 instance types
  - Mix and match in same cluster
- Disk type
  - GP3 or IO2
  - IOPS
  - GP3 - Up to 16,000
  - IO2 - Up to 64,000
- Storage
  - 50 to 16,000 GB per node
  - Autoscaling by default

Instance sizes will vary by CSP

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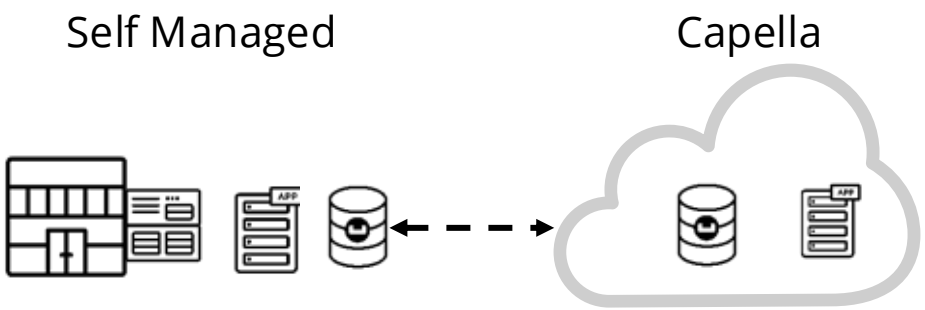
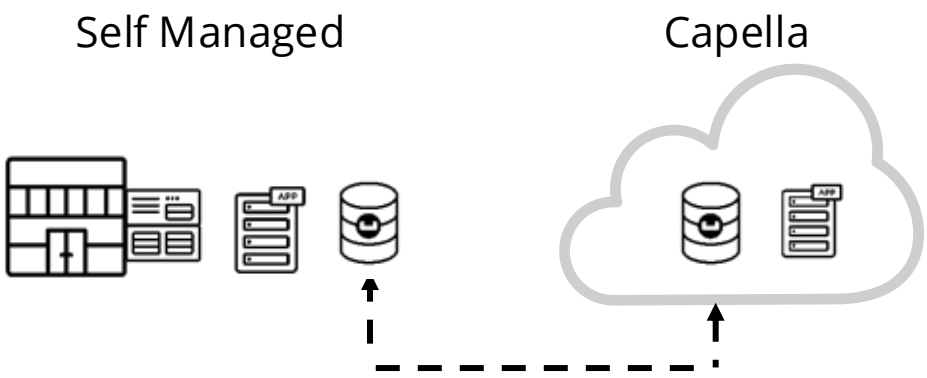
Migration and Syncing

Hybrid: Move data between self managed Server and Capella





Replication

- Configure via the Capella Control Plane
- Sync data via cross datacenter replication

Import Data

- Export data from self managed
- Import & restore data via the Control Plane

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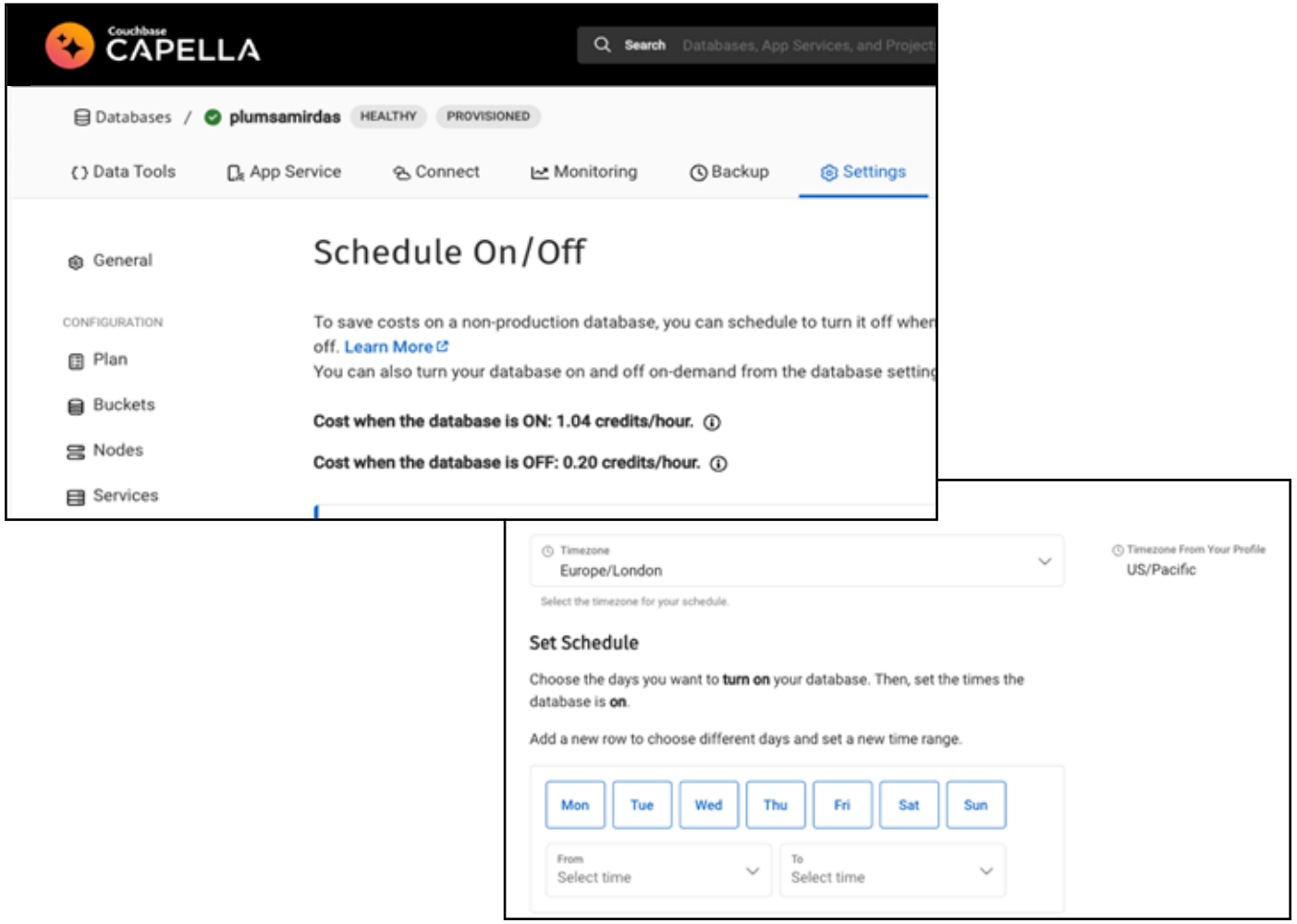
Turning Database On/Off

Reducing development credit usage

- Administrator can turn Off/On
  - Manually or scheduled
  - In the UI
  - Via an API
- When off, cost is for storage only
- App Services
  - Auto turned off (or on) with the linked cluster



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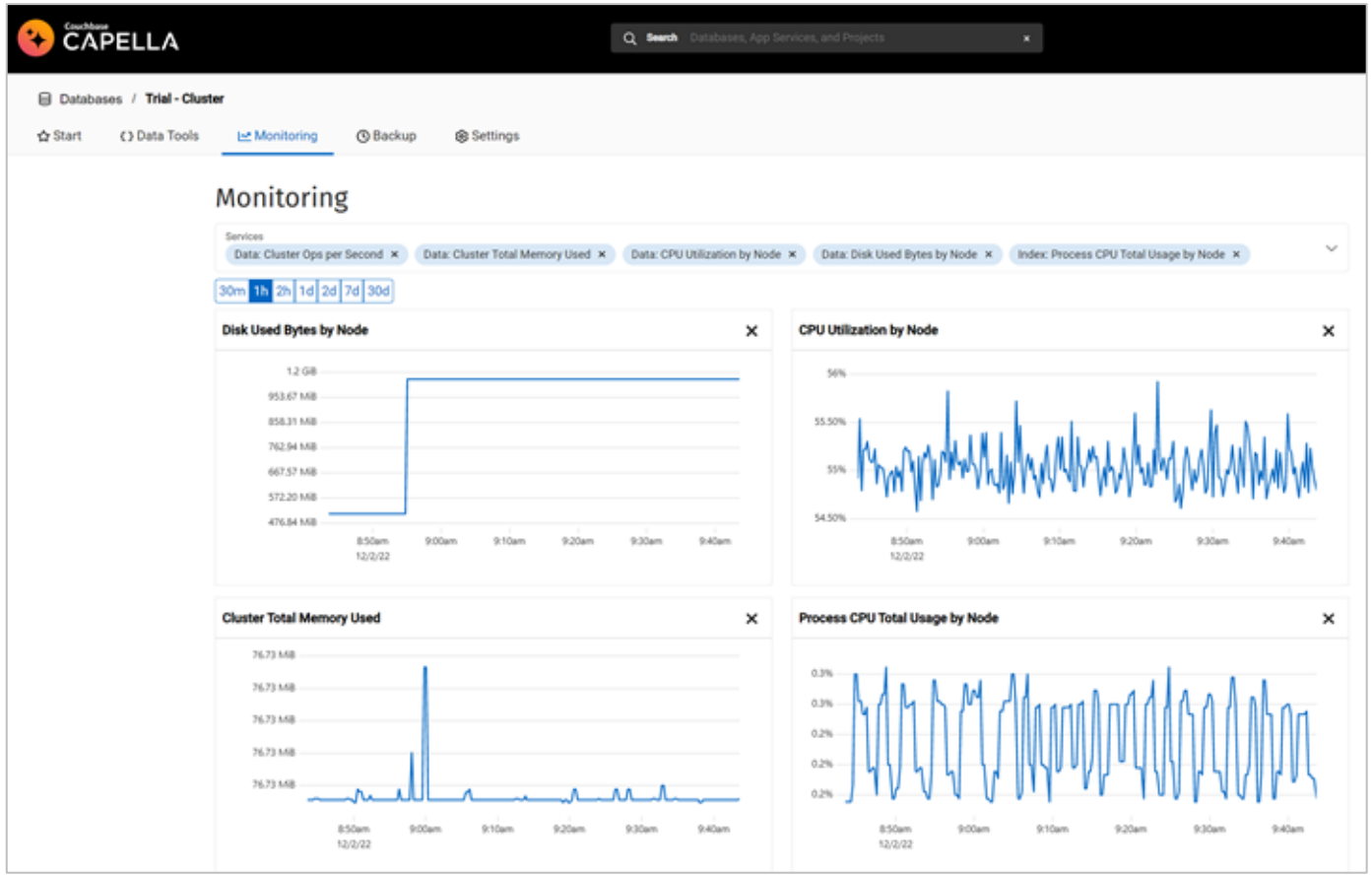
Performance Monitoring Dashboard

Comprehensive and real-time visibility

- Monitoring and alerting
- Broad metrics of status
- Database and infrastructure
- Performance threshold notifications
- Take action with one-click scale-up/out



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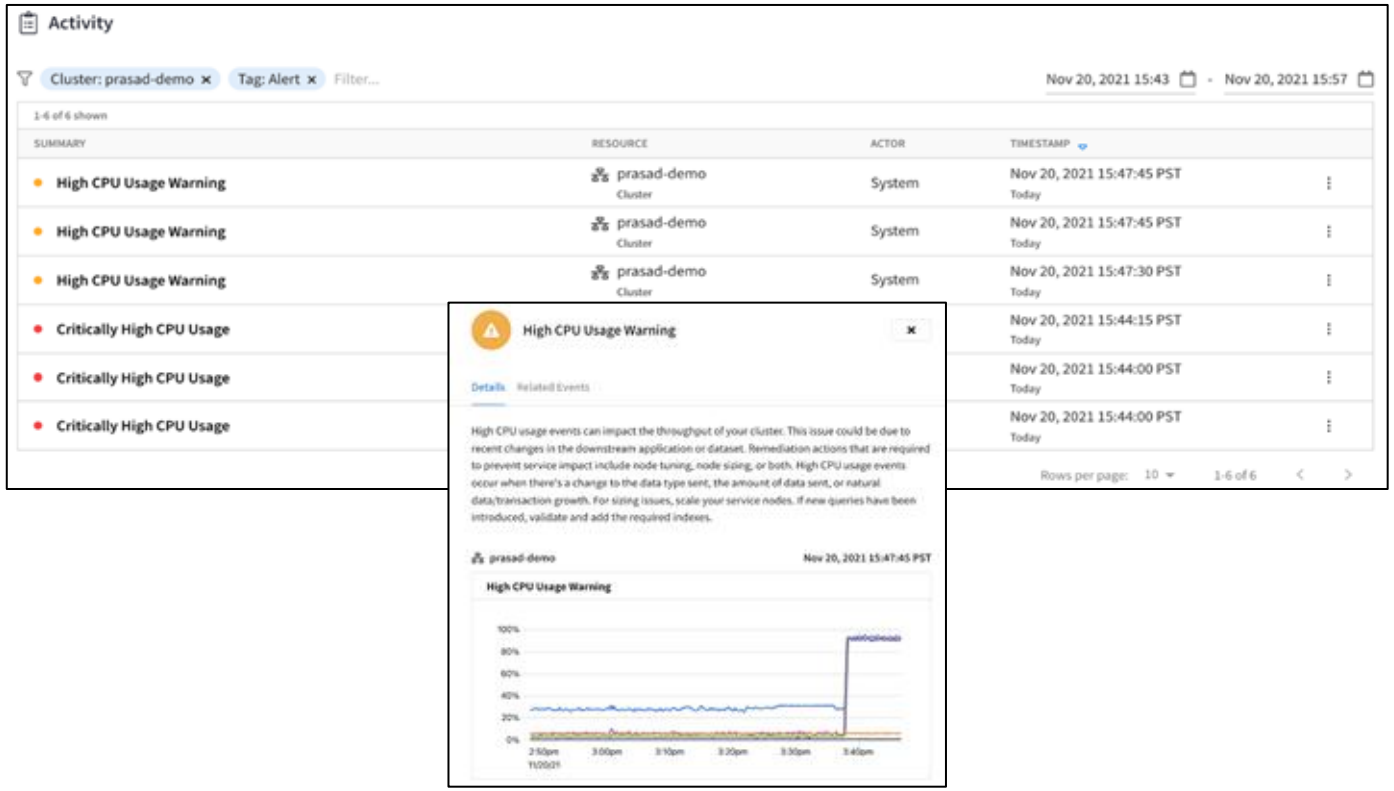
Monitoring: Alerts and Actions

Alerts make optimization and evolution easier

- Alerts
  - Severity level
    - Info
    - Warning
    - Critical
  - Recommended actions
- Webhooks supported



Recommended actions → Scale Out

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85+ Regions Supported

aws Google Cloud Microsoft Azure

North America

AWS
- US East (Ohio)
- US East (N. Virginia)
- US West (Oregon)
- Canada (Central)
- Mexico City (Mexico)

Azure
- Canada Central (Toronto)
- East US (Virginia)
- East US 2 (Virginia)
- West US 2 (Washington)
- West US 3 (Arizona)
- Central US (Iowa)
- South Central US (Texas)

Google
- Us-Central1 (Iowa)
- us-South1 (Dallas)
- Us-East1 (S. Carolina)
- Us-East4 (Virginia)
- Us-East5 (Columbus)
- Us-West1 (Oregon)
- Us-west3 (Utah)
- Us-West4 (Neada)
- Canada (Montreal)
- Canada (Toronto)

AWS
- Brazil (Sao Paulo)

Azure
- Brazil (Sao Paulo)

Google
- Sa-East1 (Sao Paulo)
- Sa-West1 (Chile)

EMEA

Azure
- EU (Netherlands)
- EU (London)
- Europe (Sweden)
- EU (Frankfurt)
- EU (Ireland)
- EU (Oslo)
- EU (Zürich)
- EU (Madrid)
- EU (Paris)
- ME (Dubai)

AWS
- EU (Frankfurt)
- EU (Ireland)
- EU (London)
- EU (Milan)
- EU (Paris)
- EU (Stockholm)
- EU (Zurich)
- ME (Bahrain)
- ME (Tel Aviv)
- ME (UAE)
- Africa (Cape Town)

Google
- Middle East (Tel Aviv)
- Middle East (Dammam)
- Europe-North1 (Finland)
- Europe-West1 (Belgium)
- Europe-West2 (London)
- Europe-West3 (Frankfurt)
- Europe-West4 (Netherlands)
- Europe-West6 (Zurich)
- Europe-West8 (Milan)
- Europe-west9 (Paris)
- Europe-central2 (Warsaw)
- Europe-southwest1 (Madrid)
- Africa (Johannesburg)

Asia Pac: AWS
- AP (Hong Kong)
- AP (Hyderabad)
- AP (Jakarta)
- AP (Melbourne)
- AP (Mumbai)
- AP (Seoul)
- AP (Singapore)
- AP (Sydney)
- AP (Thailand)
- AP (Tokyo)

Asia Pac: Azure
- AP (New South Wales)
- AP (Hong Kong)
- AP (Seoul)
- India (Pune)

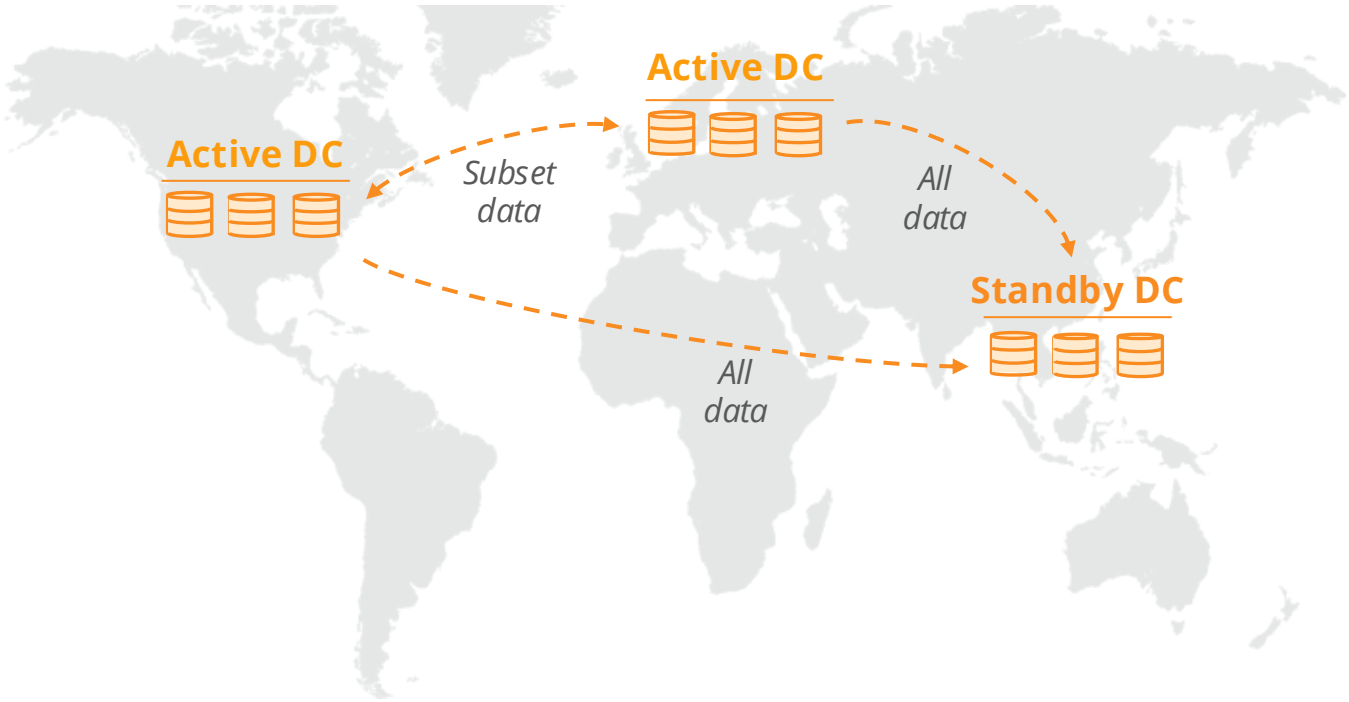
Asia Pac: Google
- Asia-East1 (Taiwan)
- asia-east2 – Hong Kong
- Asia-Northeast1 (Tokyo)
- Asia-northeast2 (Osaka)
- Asia-northeast3 (Seoul)
- Asia-South1 (Mumbai)
- Asia-South2 (New Delhi)
- Asia-Southeast1 (Singapore)
- Asia-Southeast2 (Jakarta)
- Aus-Southeast1 (Sydney)
- Aus-Southeast2 (Melbourne)

https://docs.couchbase.com/cloud/clouds/cloud-providers.html

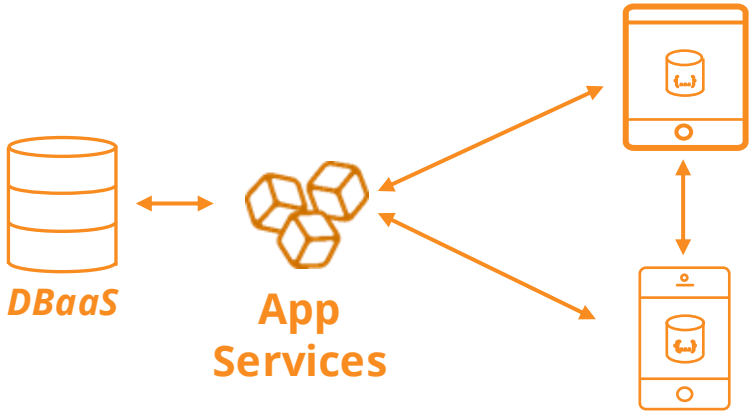
Low Latency Everywhere

Globally and to the Edge

Geo-Replication Service



Mobile Sync



- Low latency to the Edge
- Peer–to-peer syncing

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Mobile: App Services

App Services

Sync, store, query, search, and analyze at the Edge

Easier to Build & Manage

- For enterprises & mobile app developers
  - Build fast and resilient mobile apps
  - Seamlessly sync: Cloud to devices
  - Control Plane UI for configuration
- Proven offline first solution
  - Couchbase Mobile since 2014
  - Peer2peer syncing
  - Automatic data conflict resolution
  - Authenticate and manage mobile/edge users

Benefits

- Fully managed, streamlining operations
  - Set up
  - Configuration and upgrades
  - Ongoing DB services
- Faster time to market
  - New apps or extending non-mobile app
- Reduce risk with improved reliability
  - Data always available, never lost or corrupted
- Lower TCO
  - Scale with industry-leading price-performance

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What Does it Enable?

Offline-first Applications

Field Services

File claim reports, update customer policy information on mobile devices even with no network connectivity

Utility Workers

Access and update task lists, infrastructure maps, inspection reports, etc on handheld devices even with no network connectivity

Restaurants

Expedite ordering process and reduce wait times with order entry kiosks in fast food restaurants

Mobile Clinics

Access & update patient health records in rural areas even with no network connectivity

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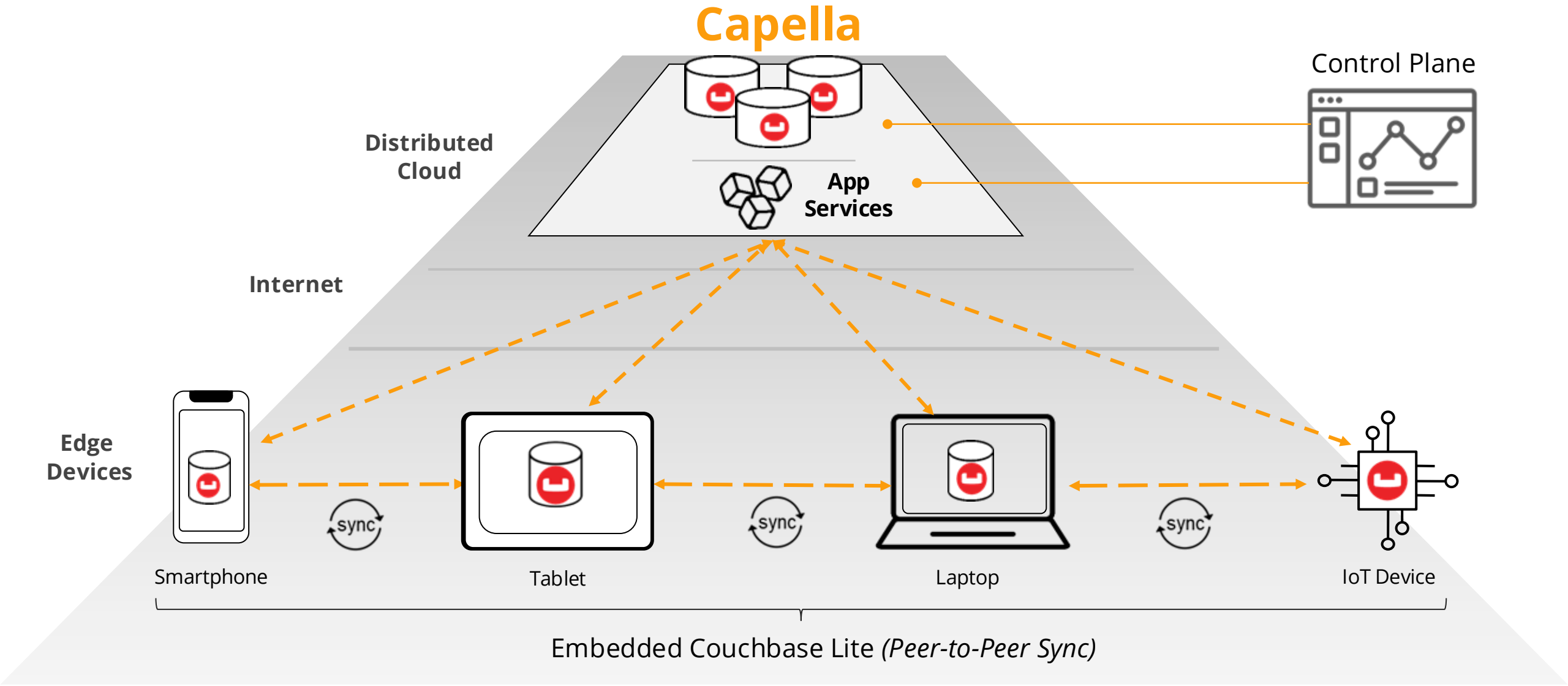
Architecture Overview



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AI Services

Billion-Scale Vector Search

Choose based on workload priorities

Search flexibility — Data volume — Accuracy — Cost efficiency — Performance

Search Vector Index

Use case:
Hybrid Search

Composite Vector Index

Use case:
Pre-filtered Search

Hyperscale Vector Index

Use case:
RAG, Agents, & Recommendations

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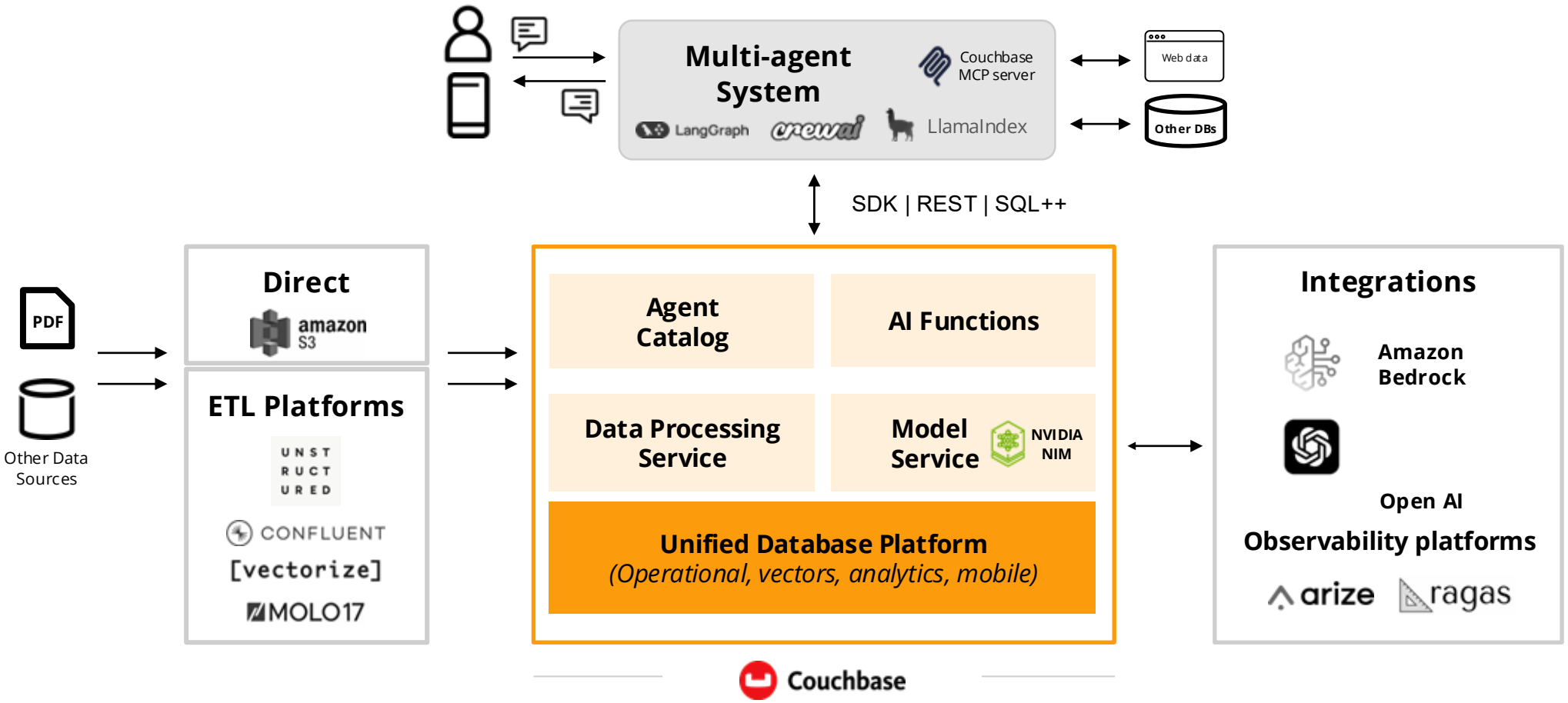
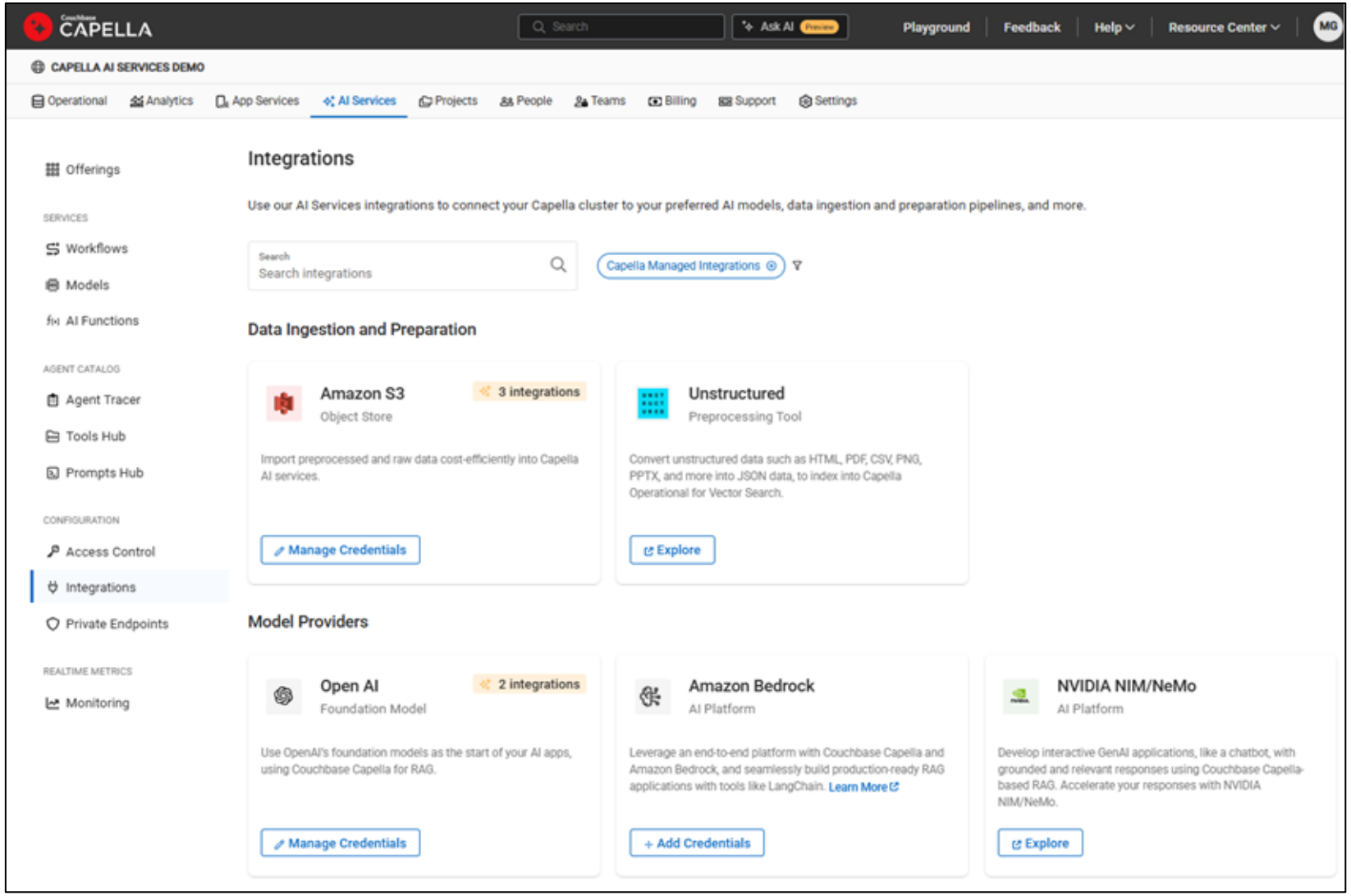
Integrated AI Services



Eliminate ETL mess, context sprawl, security risks and slow inferencing!

AI Services | **Models**: Integrated or Hosted (NVIDIA)

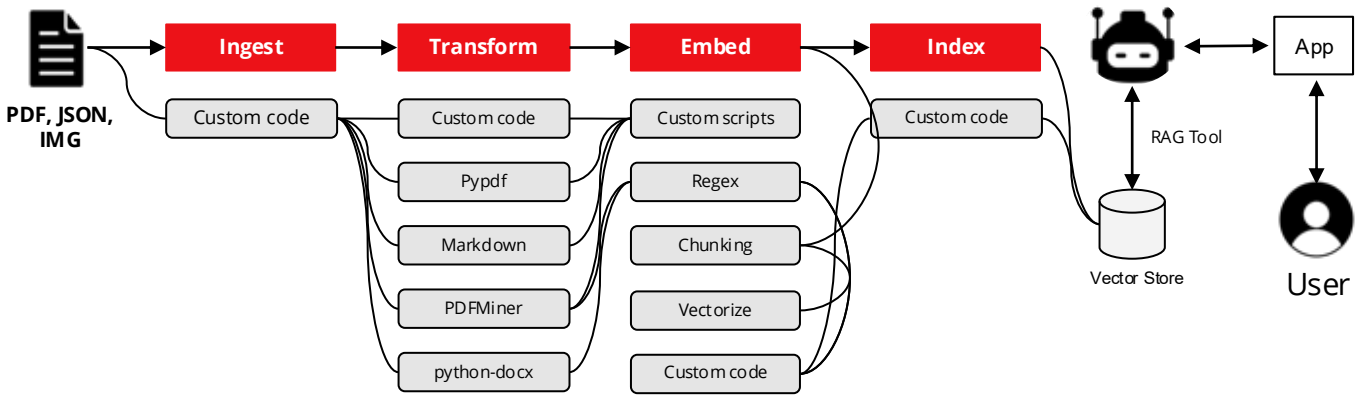


NVIDIA

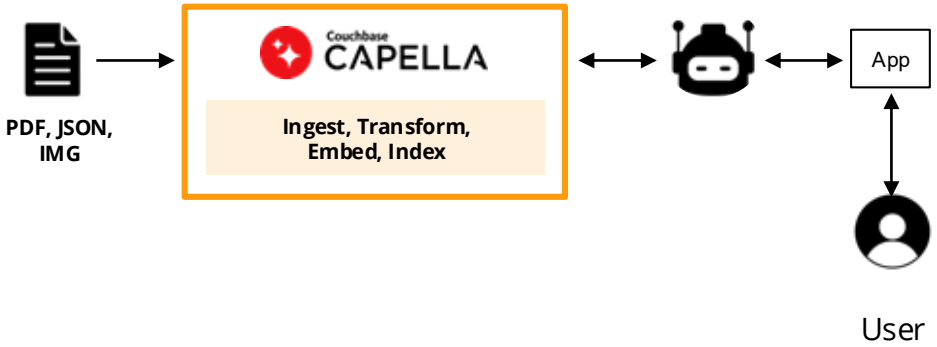
- **Hosted Models**
- **Optimized performance**
- **Built-in Guardrails**
- **Built-in Cache** (Semantic + Exact Match)

AI Services | **Data Processing & Vectorization**: Automation

DIY pipelines: are complex and require dedicated resources to maintain and improve.



Couchbase automated pipelines: eliminate low impact, undifferentiated work



Build and deploy data pipelines <u>in minutes, not days or weeks</u>

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Agent Catalog: Easing Deployment & Governance

Challenges	Agent Catalog
Tool Explosion	Tool Hub
Inconsistent Agent Behavior	Prompt Hub
Lack of Traceability	Agent Tracer

Benefits

- **Tool reusability** accelerates agent development time
- **Tool selectivity** improves accuracy and reduces costs
- **Prompt versioning** allows consistent agent behavior
- **Traceability** helps root cause problematic agent sessions

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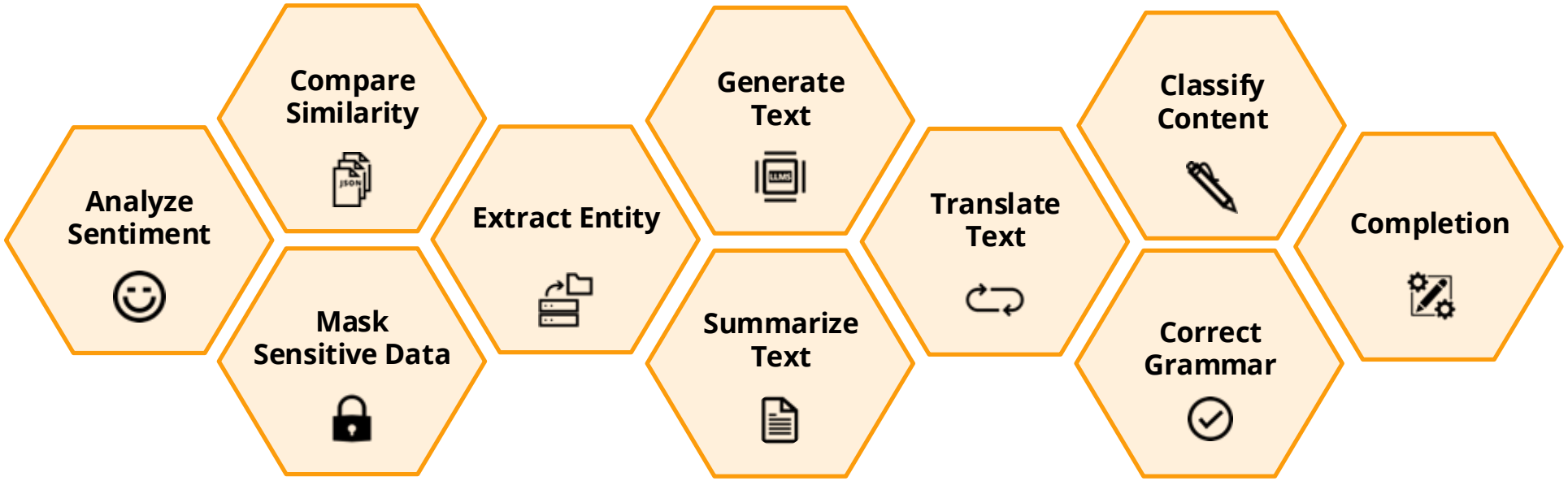
AI Services | **AI Functions:** Enhance Data via SQL++



Easily invoke LLMs using SQL++

Improve responses

Secure: Avoid internet traversal

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Security

Multi-layered, End-to-End Protection

Infrastructure Security

- Hardened environment
- Protected OS images
- Automated configuration

Network Security

- Default no ingress
- VPC peering
- AWS,GCP, Azure Private Networking

Access Security

- Control & Data Planes
- Application access
- Authentication / Authorization

Data Security

- Data in transit & rest
- Field level encryption
- Backup & restore

Vulnerability Management

- Threat detection
- Logging & alerting
- Security processes

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Capella Security: Compliance Achievements

- Capella data protection
  - **Data Isolation**
    - Single-tenant and dedicated clusters
  - **Access protection**
    - Rigid authentication and authorization controls
  - **Locked down**
    - Data plane infrastructure
  - **Minimization**
    - Secured with narrowed surface area
  - **Intrusion detection**
    - Real-time managed
  - **Rigorous security practices**
    - SOC 2, HIPAA, PCI-DSS, ISO27001 and more

https://www.couchbase.com/products/capella/trust/

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Pricing Model

Capella Credits

Credits are the currency of Couchbase Capella

Benefits

- Discounts available
- Flexible usage throughout term
- Additional multi-year savings
- Up to 12-month expiration
- Opportunity to extend when consumed

Sizing estimates available to forecast credit requirements

Capella Services Summary

All three service plans consume credits at the same rate

Product/Feature	Variables	Price Meter(s)	Example
Database	• Node count • vCPU & RAM per node • SSD (size & type) per node • Cloud vendor/region	• Running Instance-hr • Stopped Instance-hr	• 0.98 credits per running instance/hr • 0.10 credits per stopped instance/hr
Backup	• Daily backup volume size • Cloud vendor/region	• GiB-day	• 0.065 credits per GiB/month
App Services	• Node count • vCPU & RAM per node • GiB processed by App Services • Cloud vendor/region	• Instance-hr • GiB processed/hr > 1GiB/hr	• 0.82 credits per instance/hr • 0.045 credits per GiB processed/hr above 1GiB
Data Transfer	• Cloud vendor charges incurred for inter-zone, cross-region & internet egress/ingress	• Cloud vendor IaaS cost	• $10 in data transfer/day = 10 credits/day
Private Endpoints	• Cloud vendor charges incurred for hosting private endpoint (PE) infrastructure	• Cloud vendor IaaS cost (network load balancers)	• $10 in PE IaaS/day = 10 credits/day

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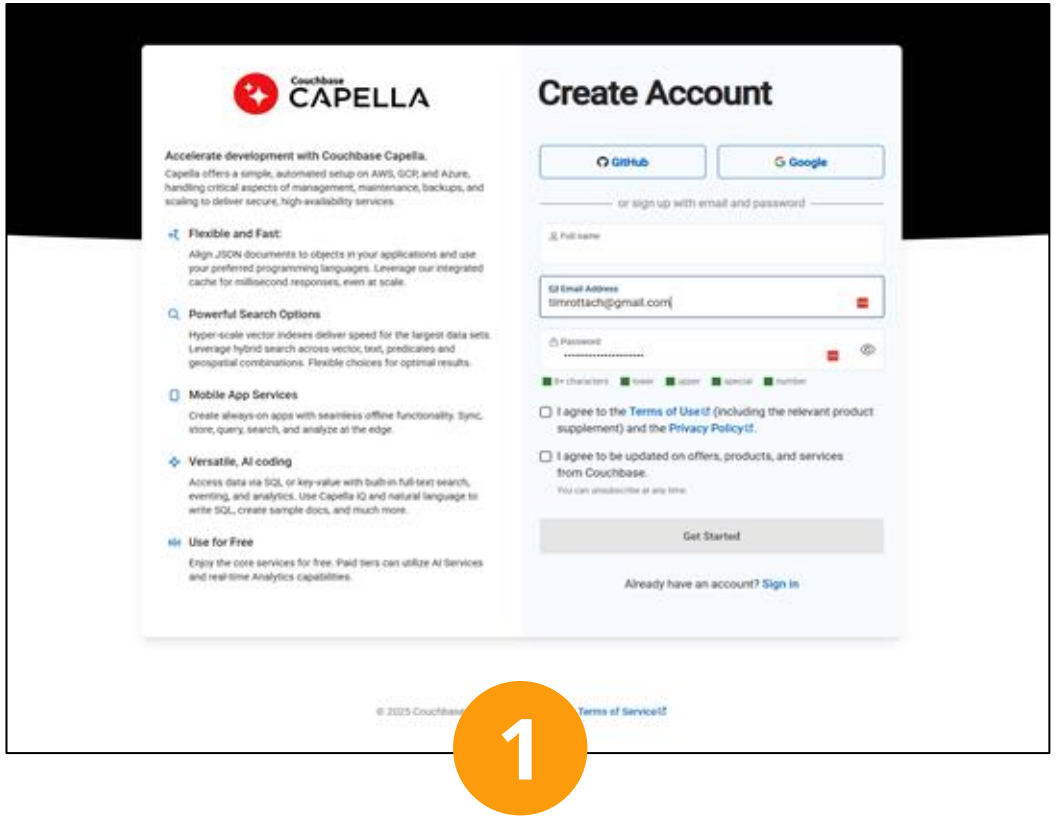
Free Tier

Be up and running in under 2 minutes



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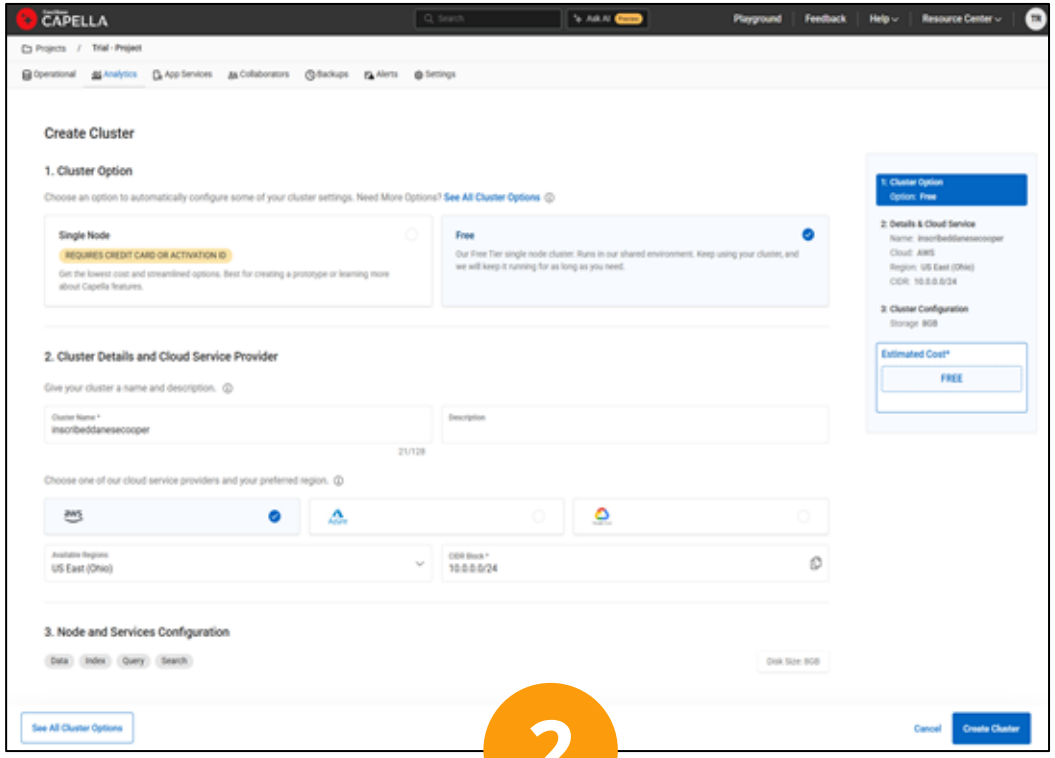

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